

Wk5-Computing Simple Statistics

May 7, 2023

```
[1]: import nltk
nltk.download('all')
```

```
[nltk_data] Downloading collection 'all'
[nltk_data] |
[nltk_data] | Downloading package abc to /home/jovyan/nltk_data...
[nltk_data] | Package abc is already up-to-date!
[nltk_data] | Downloading package alpino to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package alpino is already up-to-date!
[nltk_data] | Downloading package averaged_perceptron_tagger to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package averaged_perceptron_tagger is already up-
[nltk_data] | to-date!
[nltk_data] | Downloading package averaged_perceptron_tagger_ru to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package averaged_perceptron_tagger_ru is already
[nltk_data] | up-to-date!
[nltk_data] | Downloading package basque_grammars to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package basque_grammars is already up-to-date!
[nltk_data] | Downloading package bcp47 to
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[nltk_data] | Downloading package biocreative_ppi to
[nltk_data] | /home/jovyan/nltk_data...
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[nltk_data] | Downloading package bllip_wsj_no_aux to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package bllip_wsj_no_aux is already up-to-date!
[nltk_data] | Downloading package book_grammars to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package book_grammars is already up-to-date!
[nltk_data] | Downloading package brown to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package brown is already up-to-date!
[nltk_data] | Downloading package brown_tei to
[nltk_data] | /home/jovyan/nltk_data...
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[nltk_data] | Package brown_tei is already up-to-date!
[nltk_data] | Downloading package cess_cat to
[nltk_data] | /home/jovyan/nltk_data...
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[nltk_data] | Downloading package cess_esp to
[nltk_data] | /home/jovyan/nltk_data...
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[nltk_data] | Downloading package chat80 to
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[nltk_data] | Downloading package city_database to
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[nltk_data] | Downloading package cmudict to
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[nltk_data] | date!
[nltk_data] | Downloading package comtrans to
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[nltk_data] | Downloading package dependency_treebank to
[nltk_data] | /home/jovyan/nltk_data...
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[nltk_data] | Downloading package dolch to
[nltk_data] | /home/jovyan/nltk_data...
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[nltk_data] | Downloading package europarl_raw to
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[nltk_data] | Package europarl_raw is already up-to-date!
[nltk_data] | Downloading package extended_omw to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package extended_omw is already up-to-date!
[nltk_data] | Downloading package floresta to

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[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package floresta is already up-to-date!
[nltk_data] | Downloading package framenet_v15 to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package framenet_v15 is already up-to-date!
[nltk_data] | Downloading package framenet_v17 to
[nltk_data] | /home/jovyan/nltk_data...
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[nltk_data] | Downloading package gazetteers to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package gazetteers is already up-to-date!
[nltk_data] | Downloading package genesis to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package genesis is already up-to-date!
[nltk_data] | Downloading package gutenber to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package gutenber is already up-to-date!
[nltk_data] | Downloading package ieer to /home/jovyan/nltk_data...
[nltk_data] | Package ieer is already up-to-date!
[nltk_data] | Downloading package inaugural to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package inaugural is already up-to-date!
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[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package indian is already up-to-date!
[nltk_data] | Downloading package jeita to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package jeita is already up-to-date!
[nltk_data] | Downloading package kimmo to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package kimmo is already up-to-date!
[nltk_data] | Downloading package knbc to /home/jovyan/nltk_data...
[nltk_data] | Package knbc is already up-to-date!
[nltk_data] | Downloading package large_grammars to
[nltk_data] | /home/jovyan/nltk_data...
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[nltk_data] | Downloading package lin_thesaurus to
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[nltk_data] | Package lin_thesaurus is already up-to-date!
[nltk_data] | Downloading package mac_morpho to
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[nltk_data] | Downloading package machado to
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[nltk_data] | Downloading package masc_tagged to
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[nltk_data] | Package masc_tagged is already up-to-date!

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[nltk_data] | Downloading package maxent_ne_chunker to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package maxent_ne_chunker is already up-to-date!
[nltk_data] | Downloading package maxent_treebank_pos_tagger to
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[nltk_data] | to-date!
[nltk_data] | Downloading package moses_sample to
[nltk_data] | /home/jovyan/nltk_data...
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[nltk_data] | Downloading package movie_reviews to
[nltk_data] | /home/jovyan/nltk_data...
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[nltk_data] | Downloading package mte_teip5 to
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[nltk_data] | Downloading package mwa_ppdb to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package mwa_ppdb is already up-to-date!
[nltk_data] | Downloading package names to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package names is already up-to-date!
[nltk_data] | Downloading package nombank.1.0 to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package nombank.1.0 is already up-to-date!
[nltk_data] | Downloading package nonbreaking_prefixes to
[nltk_data] | /home/jovyan/nltk_data...
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[nltk_data] | Downloading package nps_chat to
[nltk_data] | /home/jovyan/nltk_data...
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[nltk_data] | Downloading package panlex_swadesh to
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[nltk_data] | Package panlex_swadesh is already up-to-date!
[nltk_data] | Downloading package paradigms to
[nltk_data] | /home/jovyan/nltk_data...
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[nltk_data] | Downloading package pe08 to /home/jovyan/nltk_data...
[nltk_data] | Package pe08 is already up-to-date!
[nltk_data] | Downloading package perluniprops to

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[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package perluniprops is already up-to-date!
[nltk_data] | Downloading package pil to /home/jovyan/nltk_data...
[nltk_data] | Package pil is already up-to-date!
[nltk_data] | Downloading package pl196x to
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[nltk_data] | Downloading package porter_test to
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[nltk_data] | Downloading package ppattach to
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[nltk_data] | Package ppattach is already up-to-date!
[nltk_data] | Downloading package problem_reports to
[nltk_data] | /home/jovyan/nltk_data...
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[nltk_data] | Downloading package product_reviews_1 to
[nltk_data] | /home/jovyan/nltk_data...
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[nltk_data] | Downloading package product_reviews_2 to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package product_reviews_2 is already up-to-date!
[nltk_data] | Downloading package propbank to
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[nltk_data] | Package pros_cons is already up-to-date!
[nltk_data] | Downloading package ptb to /home/jovyan/nltk_data...
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[nltk_data] | Downloading package punkt to
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[nltk_data] | Package punkt is already up-to-date!
[nltk_data] | Downloading package qc to /home/jovyan/nltk_data...
[nltk_data] | Package qc is already up-to-date!
[nltk_data] | Downloading package reuters to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package reuters is already up-to-date!
[nltk_data] | Downloading package rslp to /home/jovyan/nltk_data...
[nltk_data] | Package rslp is already up-to-date!
[nltk_data] | Downloading package rte to /home/jovyan/nltk_data...
[nltk_data] | Package rte is already up-to-date!
[nltk_data] | Downloading package sample_grammars to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package sample_grammars is already up-to-date!
[nltk_data] | Downloading package semcor to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package semcor is already up-to-date!

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[nltk_data] | Downloading package senseval to
[nltk_data] |   /home/jovyan/nltk_data...
[nltk_data] | Package senseval is already up-to-date!
[nltk_data] | Downloading package sentence_polarity to
[nltk_data] |   /home/jovyan/nltk_data...
[nltk_data] | Package sentence_polarity is already up-to-date!
[nltk_data] | Downloading package sentiwordnet to
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[nltk_data] | Package sentiwordnet is already up-to-date!
[nltk_data] | Downloading package shakespeare to
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[nltk_data] | Package shakespeare is already up-to-date!
[nltk_data] | Downloading package sinica_treebank to
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[nltk_data] | Package sinica_treebank is already up-to-date!
[nltk_data] | Downloading package smultron to
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[nltk_data] | Package smultron is already up-to-date!
[nltk_data] | Downloading package snowball_data to
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[nltk_data] | Package snowball_data is already up-to-date!
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[nltk_data] | Package state_union is already up-to-date!
[nltk_data] | Downloading package stopwords to
[nltk_data] |   /home/jovyan/nltk_data...
[nltk_data] | Package stopwords is already up-to-date!
[nltk_data] | Downloading package subjectivity to
[nltk_data] |   /home/jovyan/nltk_data...
[nltk_data] | Package subjectivity is already up-to-date!
[nltk_data] | Downloading package swadesh to
[nltk_data] |   /home/jovyan/nltk_data...
[nltk_data] | Package swadesh is already up-to-date!
[nltk_data] | Downloading package switchboard to
[nltk_data] |   /home/jovyan/nltk_data...
[nltk_data] | Package switchboard is already up-to-date!
[nltk_data] | Downloading package tagsets to
[nltk_data] |   /home/jovyan/nltk_data...
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[nltk_data] |   /home/jovyan/nltk_data...
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[nltk_data] | Downloading package toolbox to
[nltk_data] |   /home/jovyan/nltk_data...
[nltk_data] | Package toolbox is already up-to-date!

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[nltk_data] | Downloading package treebank to
[nltk_data] |     /home/jovyan/nltk_data...
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[nltk_data] | Downloading package twitter_samples to
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[nltk_data] | Package twitter_samples is already up-to-date!
[nltk_data] | Downloading package udhr to /home/jovyan/nltk_data...
[nltk_data] | Package udhr is already up-to-date!
[nltk_data] | Downloading package udhr2 to
[nltk_data] |     /home/jovyan/nltk_data...
[nltk_data] | Package udhr2 is already up-to-date!
[nltk_data] | Downloading package unicode_samples to
[nltk_data] |     /home/jovyan/nltk_data...
[nltk_data] | Package unicode_samples is already up-to-date!
[nltk_data] | Downloading package universal_tagset to
[nltk_data] |     /home/jovyan/nltk_data...
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[nltk_data] |     date!
[nltk_data] | Downloading package vader_lexicon to
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[nltk_data] | Downloading package verbnet to
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[nltk_data] | Package verbnet is already up-to-date!
[nltk_data] | Downloading package verbnet3 to
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[nltk_data] | Package verbnet3 is already up-to-date!
[nltk_data] | Downloading package webtext to
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[nltk_data] | Package webtext is already up-to-date!
[nltk_data] | Downloading package wmt15_eval to
[nltk_data] |     /home/jovyan/nltk_data...
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[nltk_data] | Downloading package word2vec_sample to
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[nltk_data] | Package word2vec_sample is already up-to-date!
[nltk_data] | Downloading package wordnet to
[nltk_data] |     /home/jovyan/nltk_data...
[nltk_data] | Package wordnet is already up-to-date!
[nltk_data] | Downloading package wordnet2021 to
[nltk_data] |     /home/jovyan/nltk_data...
[nltk_data] | Package wordnet2021 is already up-to-date!
[nltk_data] | Downloading package wordnet2022 to
[nltk_data] |     /home/jovyan/nltk_data...
[nltk_data] | Package wordnet2022 is already up-to-date!

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```

[nltk_data] | Downloading package wordnet31 to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package wordnet31 is already up-to-date!
[nltk_data] | Downloading package wordnet_ic to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package wordnet_ic is already up-to-date!
[nltk_data] | Downloading package words to
[nltk_data] | /home/jovyan/nltk_data...
[nltk_data] | Package words is already up-to-date!
[nltk_data] | Downloading package ycoe to /home/jovyan/nltk_data...
[nltk_data] | Package ycoe is already up-to-date!
[nltk_data] |
[nltk_data] Done downloading collection all

```

[1]: True

[2]: `from nltk.book import *`

```

*** Introductory Examples for the NLTK Book ***
Loading text1, ..., text9 and sent1, ..., sent9
Type the name of the text or sentence to view it.
Type: 'texts()' or 'sents()' to list the materials.
text1: Moby Dick by Herman Melville 1851
text2: Sense and Sensibility by Jane Austen 1811
text3: The Book of Genesis
text4: Inaugural Address Corpus
text5: Chat Corpus
text6: Monty Python and the Holy Grail
text7: Wall Street Journal
text8: Personals Corpus
text9: The Man Who Was Thursday by G . K . Chesterton 1908

```

[3]: `sents()`

```

sent1: Call me Ishmael .
sent2: The family of Dashwood had long been settled in Sussex .
sent3: In the beginning God created the heaven and the earth .
sent4: Fellow - Citizens of the Senate and of the House of Representatives :
sent5: I have a problem with people PMing me to lol JOIN
sent6: SCENE 1 : [ wind ] [ clop clop clop ] KING ARTHUR : Whoa there !
sent7: Pierre Vinken , 61 years old , will join the board as a nonexecutive
director Nov. 29 .
sent8: 25 SEXY MALE , seeks attrac older single lady , for discreet encounters .
sent9: THE suburb of Saffron Park lay on the sunset side of London , as red and
ragged as a cloud of sunset .

```

[4]: `text1`

[4]: <Text: Moby Dick by Herman Melville 1851>

```
[5]: # Part 2: Counting Vocab
```

```
[6]: #All tokens in the text  
len(text1)
```

[6]: 260819

```
[7]: #Removes duplicates  
set(text1)
```

```
[7]: {'divinely',  
      'nooks',  
      'incomplete',  
      'grinding',  
      'villains',  
      'judgmatically',  
      'either',  
      'REMAINING',  
      'Cannon',  
      'confined',  
      'jobs',  
      'legatees',  
      'dumfoundered',  
      'wages',  
      'quaint',  
      'Cough',  
      'Fishes',  
      'subscribes',  
      'pastiles',  
      'Gnawed',  
      'lances',  
      'capturing',  
      'iron',  
      'opium',  
      'constrained',  
      'prodigies',  
      'eloquent',  
      'crackled',  
      'unavoidably',  
      'heron',  
      'crawled',  
      'scorch',  
      'sympathetically',  
      'spoons',  
      'dim',
```

'TERROR',
'tide',
'fidelity',
'handing',
'Woods',
'belfry',
'BALEINE',
'gurgling',
'hospitals',
'vivid',
'hideous',
'respond',
'SINGS',
'indefiniteness',
'forewarnings',
'harp',
'interested',
'Cathedral',
'fearlessness',
'upraising',
'parading',
'aged',
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[8]: #Sorted unique tokens
sorted(set(text1))

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'BRACKTON',
'BREACH',
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'CURRENTS',
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'Cherries',
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'Chowder',
'Christ',
'Christendom',
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'Christianity',
'Christians',
'Christmas',
'Church',
'Cinque',
'Circassian',
'Circumambulate',
'Cistern',
'Civitas',
'Clam',
'Clap',
'Claus',
'Clay',
'Clear',
'Clearing',

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'Cleveland',
'Clifford',
'Clinging',
'Clootz',
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'Commonly',
'Commons',
'Commonwealth',
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'Connecticut',

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'Conversation',
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'Cowper',
'Crab',
'Crack',
'Crammer',
'Crappo',
'Crappoes',
'Crazed',
'Creagh',
'Created',
'Cretan',
'Crete',
...]

```
[9]: #Counts unique tokens  
len(sorted(set(text1)))
```

```
[9]: 19317
```

```
[10]: def lexical_diversity(text):  
      return len(set(text)) / len(text)
```

```
[11]: lexical_diversity(text1)
```

```
[11]: 0.07406285585022564
```

```
[12]: #Derrick's Example  
testtext = ["apple", "banana", "cherry", "cherry"]  
print("Set returns",set(testtext)," , len(set(testtext)) returns",  
      ↪,len(set(testtext))," , len(testtext) returns ",len(testtext))  
print("So lexical_diversity is 3/4, which is",lexical_diversity(testtext))
```

```
Set returns {'banana', 'cherry', 'apple'} , len(set(testtext)) returns 3 ,  
len(testtext) returns 4  
So lexical_diversity is 3/4, which is 0.75
```

```
[13]: # Part 3: Freq distribution
```

```
[14]: f = FreqDist(text1)
```

```
[15]: f
```

```
[15]: FreqDist({' ': 18713, 'the': 13721, '.': 6862, 'of': 6536, 'and': 6024, 'a':  
4569, 'to': 4542, ';': 4072, 'in': 3916, 'that': 2982, ...})
```

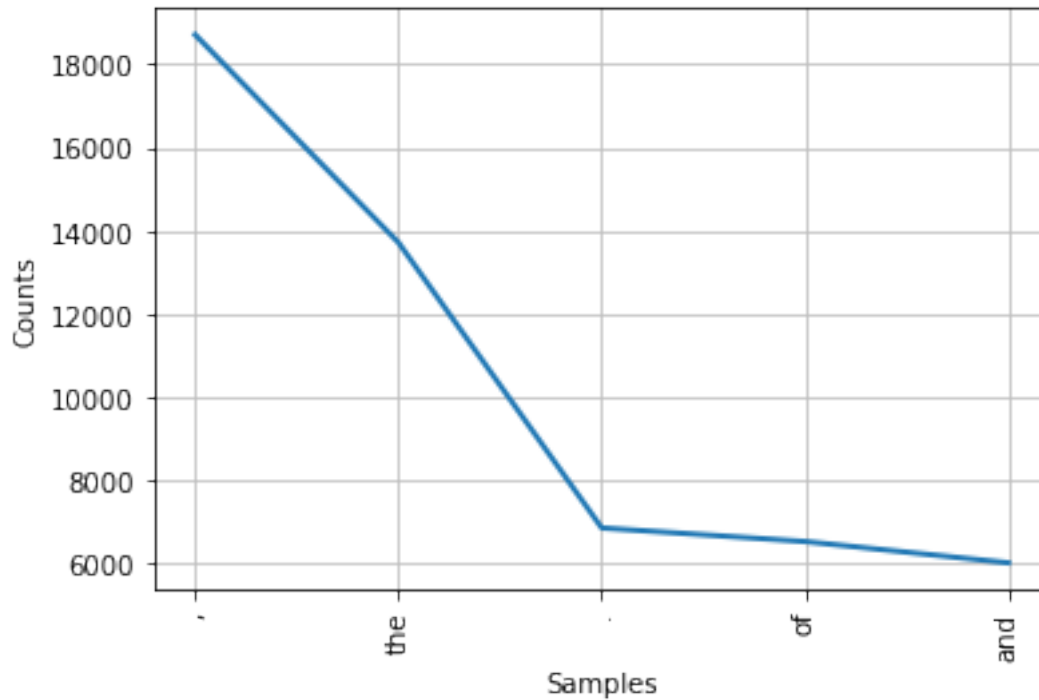
```
[16]: #5 most common terms  
print("5 most comment terms and their respective counts are",f.most_common(5))
```

```
5 most comment terms and their respective counts are [(' ', 18713), ('the',  
13721), ('.', 6862), ('of', 6536), ('and', 6024)]
```

```
[17]: print("The word and appears",f['and'], "times")
```

```
The word and appears 6024 times
```

```
[18]: f.plot(5, cumulative = False)
```



[18]: <matplotlib.axes._subplots.AxesSubplot at 0x7fc5bce64490>

```
[19]: #Derrick's Example
testtext = ["apple", "banana", "cherry", "cherry"]
testf = FreqDist(testtext)
testf
```

[19]: FreqDist({'cherry': 2, 'apple': 1, 'banana': 1})

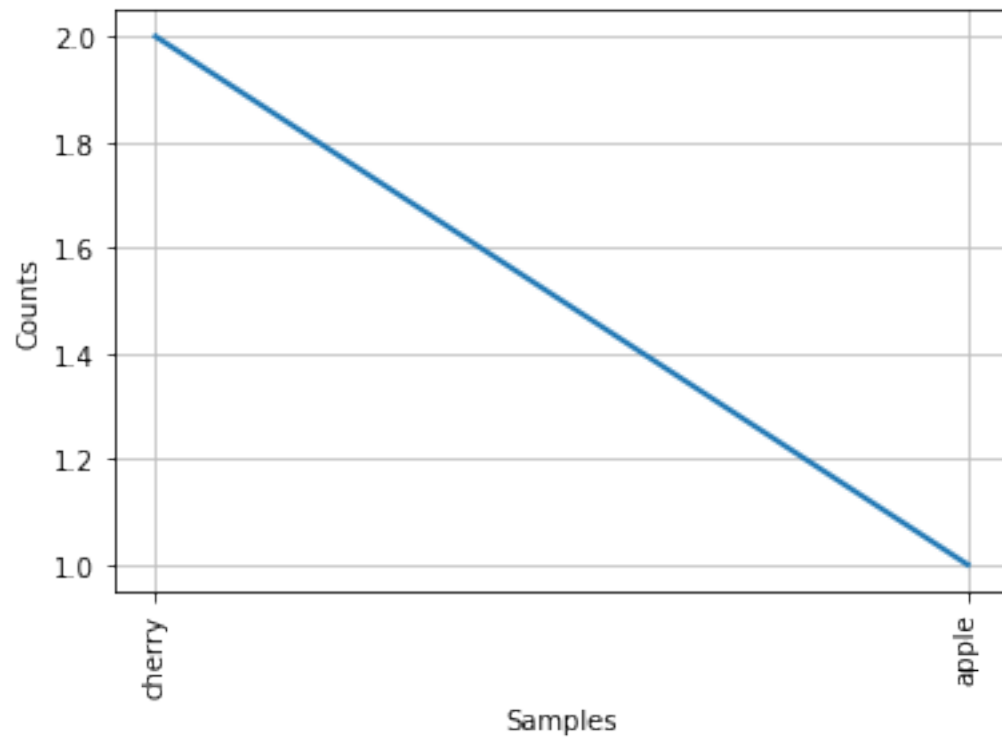
```
[20]: #2 most common terms --> Notice how Banana is cut off because its alphabetical
      ↪ order
print("2 most common terms and their respective counts are",testf.
      ↪ most_common(2))
```

2 most common terms and their respective counts are [('cherry', 2), ('apple', 1)]

```
[21]: print("The word cherry appears",testf['cherry'], "times")
```

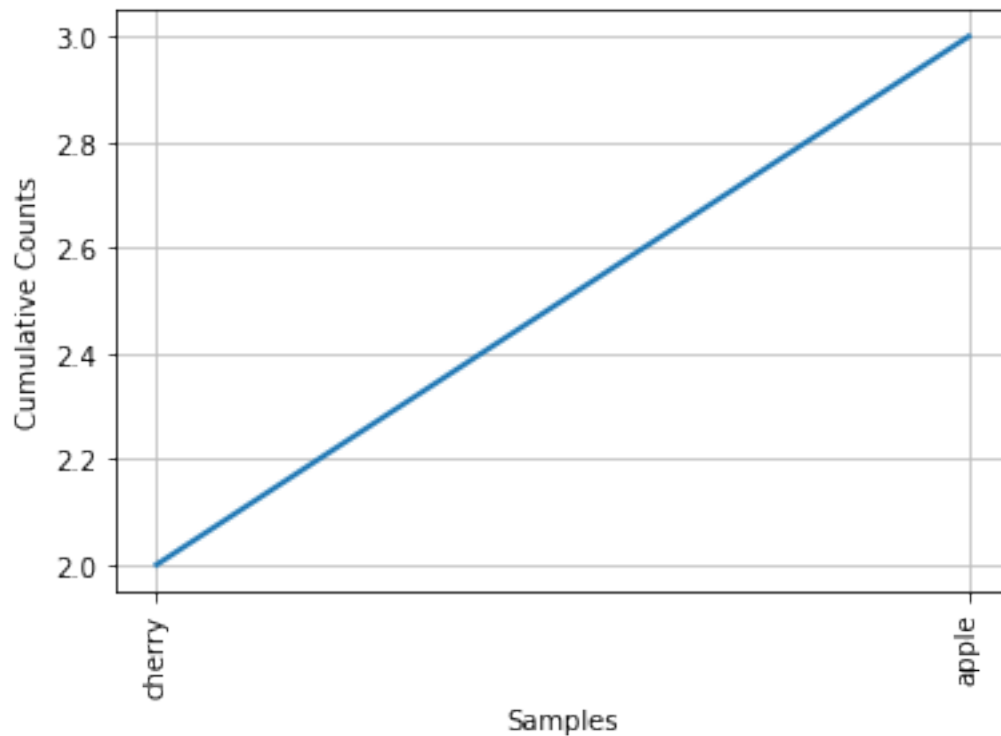
The word cherry appears 2 times

```
[22]: testf.plot(2, cumulative = False)
```



[22]: <matplotlib.axes._subplots.AxesSubplot at 0x7fc5ba989dd0>

```
[23]: testf.plot(2, cumulative = True)
```



[23]: <matplotlib.axes._subplots.AxesSubplot at 0x7fc5ba8f8bd0>

[24]: *#Part 4 - Collocations*

[25]: `text1.collocations()`

Sperm Whale; Moby Dick; White Whale; old man; Captain Ahab; sperm whale; Right Whale; Captain Peleg; New Bedford; Cape Horn; cried Ahab; years ago; lower jaw; never mind; Father Mapple; cried Stubb; chief mate; white whale; ivory leg; one hand

[26]: *#Derrick's Example*
#Recall text1 title is Moby Dick by Herman Melville 1851 so we should have
→ 1,4,4,2,6,8,4...
`[len(w) for w in text1]`

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```
[27]: f = FreqDist([len(w) for w in text1])
```

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[28]: f
```

```
[28]: FreqDist({3: 50223, 1: 47933, 4: 42345, 2: 38513, 5: 26597, 6: 17111, 7: 14399,  
8: 9966, 9: 6428, 10: 3528, ...})
```

```
[29]: #Derrick's Example  
#testtext = ["apple", "banana", "cherry", "cherry"]  
testf1 = FreqDist([len(w) for w in testtext])
```

```
[30]: #Derrick's Example  
testf1
```

```
[30]: FreqDist({6: 3, 5: 1})
```

```
[ ]:
```